VERITRUST

Link Detection Implementation Handbook

HF-managed production revision

ARCHITECTURE	HF-MANAGED	EVIDENCE-FIRST	PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: Architecture_Source_of_Truth.yaml; Model_Registry_Proposal.json; HuggingFace_Deployment_Matrix.csv; Implementation_Backlog.csv; Database_Target_Contract.md; Verification_Test_Plans.md; Migration_Roadmap.md.

Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability and calibration_version remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Executive decision and P0 semantic defect

Block promotion until the mapping is corrected and verified through unit, provider-golden and historical replay tests.

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The current Swift normalizer maps URLBERT `LABEL_1` to phishing and `LABEL_3` to defacement. The immutable model card at revision 917ded0 states `LABEL_1=defacement` and `LABEL_3=phishing`.

2. Current path and target stages

Current parsing uses a last-two-label base-domain approximation, calls a mutable Swift model, may fall back to another provider/local heuristic and combines model/rule scores using fixed weights. The target uses independent deterministic, learned and live-enrichment evidence.

```
RAW URL -> BOUNDED PARSE -> CANONICAL IDENTITY -> STATIC EVIDENCE
-> PERSIST REGISTRY RESOLUTION -> HF URLBERT ENDPOINT
-> OPTIONAL SSRF-SAFE ENRICHMENT / PAGE-TEXT HF ENDPOINT
-> EVIDENCE SUFFICIENCY -> GATEWAY POLICY
```

3. URL identity: PSL, IDNA and confusables

Preserve raw URL and generate separately versioned display, comparison and model representations. Use a maintained Public Suffix List for registrable domains, IDNA for host identity, Unicode confusable evidence and normalized percent-encoding without over-decoding.

- No canonicalization step may erase evidence.
- Userinfo, ports, IP literals, IPv6, encoded separators, dot variants and mixed script need explicit reason codes.
- The model input string is versioned and reproducible.

4. Static evidence and model roles

Family	Owner	Meaning
Lexical/structural	VeriTrust deterministic	length, entropy, host/path/query, redirects, confusable indicators
URLBERT	HF Endpoint	four-class learned URL evidence
Feature GBDT	private HF Endpoint	learned structured evidence; never local
ONNX shadow	HF custom Endpoint	research-only lexical evidence
Page text	private HF Endpoint	optional sanitized page-content evidence
Screenshot/brand	SigLIP2 HF Endpoint + curated index	supporting embedding similarity; never a phishing label

5. Screenshot and brand-similarity evidence

Visual similarity is supporting evidence only. It must never be converted directly into PHISHING=TRUE.

The isolated browser creates a bounded screenshot artifact. SigLIP2 at revision 75de2d5 runs on an HF Endpoint and returns an embedding. VeriTrust compares it with a curated, versioned, tenant/policy-aware, provenance-tracked and poisoning-protected reference index. Evidence records model/preprocessing, screenshot identity, index version, nearest references, similarities, threshold/calibration version, status/reasons and receipt.

6. Live enrichment and SSRF boundary

Application DNS and a URL allowlist are not enough. The network worker needs independent resolver/IP validation and egress controls.

DNS, RDAP, TLS, redirect, HTTP metadata, reputation and browser evidence are optional connectors. Resolve and classify every destination before connect; pin the approved IP per hop; revalidate redirects; prohibit private/link-local/metadata/onion/local schemes; bound bytes/time/redirects.

7. Browser sandbox

Use an isolated disposable browser identity with no tenant cookies or credentials, strict egress, no downloads, no external password manager and bounded screenshot/DOM/form metadata. Browser evidence is stored as sanitized facts and hashes; active content is never returned to clients.

- Fresh sandbox per job.
- Network and storage destroyed after retention window.
- Every redirect and subresource follows policy.

8. Contracts, label map, registry and receipt

URLBERT's exact label map is bound to revision 917ded0. Unknown labels fail. Model execution uses the canonical model string, exact tokenizer/preprocessing and endpoint deployment from a persisted registry resolution.

Label	Meaning
LABEL_0	benign
LABEL_1	defacement
LABEL_2	malware
LABEL_3	phishing

Receipt	Fields
inference_receipt.v2 fields	receipt_id, registry_resolution_id, deployment_id, deployment_bundle_id, request_id, provider, endpoint_identity, endpoint_deployment_revision, request_sha256, response_sha256, provider_request_id, components, attempt, latency_ms, billed_units, completion_status, received_at

Receipt	Fields
components[] fields	component_role, model_key, repository_id, immutable_revision, artifact_or_container_digest, preprocessing_version, input_schema_version, output_schema_version, execution_order, qualification_status
Compatibility	A valid v1 receipt is exposed as a v2 receipt with one synthetic component derived from its singular repository/model/revision/preprocessing fields; missing historical provenance remains explicit and is never fabricated.

9. Scoring, calibration and policy

Model-card F1 is not a VeriTrust production measurement.

Do not use the current 0.62/0.38 model/rule mixture as probability. Calibrate model roles independently; deterministic observations remain observations. Policy uses sufficiency, quality, connector freshness, model evidence and context. A stale reputation miss is not benign.

10. Benchmark and adversarial tests

- Temporal/domain-family split; no registrable-domain leakage.
- Benign long/encoded/IDN URLs, punycode homographs, shorteners, redirectors, login/payment contexts.
- Phishing, defacement, malware and clean subclasses with stable taxonomy.
- Parser differential tests, SSRF/DNS rebinding, redirect loops, decompression/response bombs.
- Calibration, abstention, slice errors, connector outages and cost.

11. Failure, cost, scaling and observability

URL fast path uses a protected HF endpoint with min replica only after measured SLO/cost. Enrichment and page-text are asynchronous when deadlines require. Deduplicate by canonical artifact/contract while preserving raw provenance.

- Observe label distribution, unknown-label rejection and defacement/phishing regression.
- Measure PSL/IDNA parser errors, connector freshness, sandbox failures and endpoint cold starts.

Failure	Behavior
HF fully unavailable	High-impact: fail closed or mandatory review. Advisory: degraded/uncertain. Queue only when policy and deadline permit.
One endpoint unavailable	Retry only the same immutable deployment within budget; never silently change model/provider.
Malformed/unknown output	Reject schema/label; persist failed evidence; never guess from text or label digits.
Missing calibration	May display uncalibrated evidence; cannot authorize allow/block.
Timeout/cancel	Persist terminal status and preserve partial evidence; no invented benign value.

12. API, deployment, rollout and backlog

Current `scan link`/direct/Gateway clients remain compatible. Proposed evidence endpoints expose raw/normalized/calibrated separation. Deploy URLBERT to HF Endpoint, shadow connectors/private models, then tenant canary and controlled promotion.

- VT2-001 mapping; VT2-006 identity; VT2-017 Endpoint; VT2-027/028/029 optional learned roles; VT2-034 connectors.
- Done means correct semantics, SSRF-safe evidence, exact HF receipts and no local learned inference.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.